LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Performing Butt/Lap joint by Arc welding procedure and Inspection of welded job
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Performing Butt/Lap joint by Arc welding procedure and Inspection of welded job
5. Specific Objectives	31.1.4.1 Trainee will be able to explain arc welding and brazing procedure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Butt and Lap joint preparation, Arc welding execution, and visual inspection of weld quality with instructor supervision on safety protocols.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the two primary types of arc welding joints discussed in this lesson?
(b) Define the term 'weld penetration' in the context of butt joints.
(c) List three essential safety precautions required before starting an arc welding operation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and principles of Arc Welding • Preparation of Butt and Lap joints • Selection of electrode and current settings • Step-by-step welding procedure execution • Common defects in arc welded joints	L & Ds	PPT, IB	15 min
(b)	• Execution of Butt joint welding procedure • Execution of Lap joint welding procedure • Visual inspection criteria for weld quality • Non-destructive testing methods overview • Identification and rectification of weld defects	L & Ds	PPT, PDF	20 min
(c)	• Final inspection and quality assessment • Documentation of weld parameters • Comparison of Butt vs Lap joint applications • Review of safety shutdown procedures • Summary of key learning points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference in preparation requirements between a Butt joint and a Lap joint.
(b) Describe the visual indicators that suggest a weld has insufficient penetration.
(c) Outline the correct sequence for shutting down an arc welding machine safely.

12. Home Assignment and Follow Up	Prepare a brief report comparing the structural advantages of Butt joints versus Lap joints in aircraft structural applications, referencing the master syllabus for LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Equipment setup, tools and spares for Gas welding and Brazing procedure	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min